

# Source code

## TEAM MEMBERS:

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```
// SEA SENTRY
```

```
// Arduino Mega 2560 + HC-SR04 + RYG LED + Passive Buzzer
```

```
const int trigPin = 7;
```

```
const int echoPin = 8;
```

```
const int greenLED = 30;
```

```
const int yellowLED = 31;
```

```
const int redLED = 32;
```

```
const int buzzer = 33;
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  pinMode(greenLED, OUTPUT);
```

```
  pinMode(yellowLED, OUTPUT);
```

```
pinMode(redLED, OUTPUT);
```

```
pinMode(buzzer, OUTPUT);
```

```
Serial.println("SEA SENTRY STARTED");
```

```
}
```

```
void loop() {
```

```
    // Send ultrasonic pulse
```

```
    digitalWrite(trigPin, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(trigPin, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(trigPin, LOW);
```

```
    // Wait for echo
```

```
    long duration = pulseIn(echoPin, HIGH, 30000);
```

```
    // -----
```

```
    // SENSOR FAILURE
```

```
    // -----
```

```
    if (duration == 0) {
```

```
        digitalWrite(greenLED, LOW);
```

```
digitalWrite(yellowLED, LOW);
```

```
digitalWrite(redLED, LOW);
```

```
Serial.println("!!! SENSOR FAILURE !!!");
```

```
// Special error sound
```

```
tone(buzzer, 500);
```

```
delay(200);
```

```
noTone(buzzer);
```

```
delay(100);
```

```
tone(buzzer, 500);
```

```
delay(200);
```

```
noTone(buzzer);
```

```
delay(100);
```

```
tone(buzzer, 300);
```

```
delay(500);
```

```
noTone(buzzer);
```

```
delay(1000);
```

```
}
```

```
// -----
```

```
// SENSOR WORKING
// -----
else {

    float distance = duration * 0.0343 / 2;

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    // Turn LEDs and buzzer OFF first
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, LOW);
    noTone(buzzer);

    // GREEN - SAFE
    if (distance > 20) {

        digitalWrite(greenLED, HIGH);

    }

    // YELLOW - WARNING
    else if (distance >= 10 && distance <= 20) {

        digitalWrite(yellowLED, HIGH);
```

```
tone(buzzer, 1000);
```

```
delay(200);
```

```
noTone(buzzer);
```

```
delay(500);
```

```
}
```

```
// RED - DANGER
```

```
else if (distance > 0 && distance < 10) {
```

```
digitalWrite(redLED, HIGH);
```

```
tone(buzzer, 2000);
```

```
delay(100);
```

```
noTone(buzzer);
```

```
delay(100);
```

```
}
```

```
}
```

```
delay(100);
```

```
}
```